Modeling HBV Capsid Assembly at the Residue Scale

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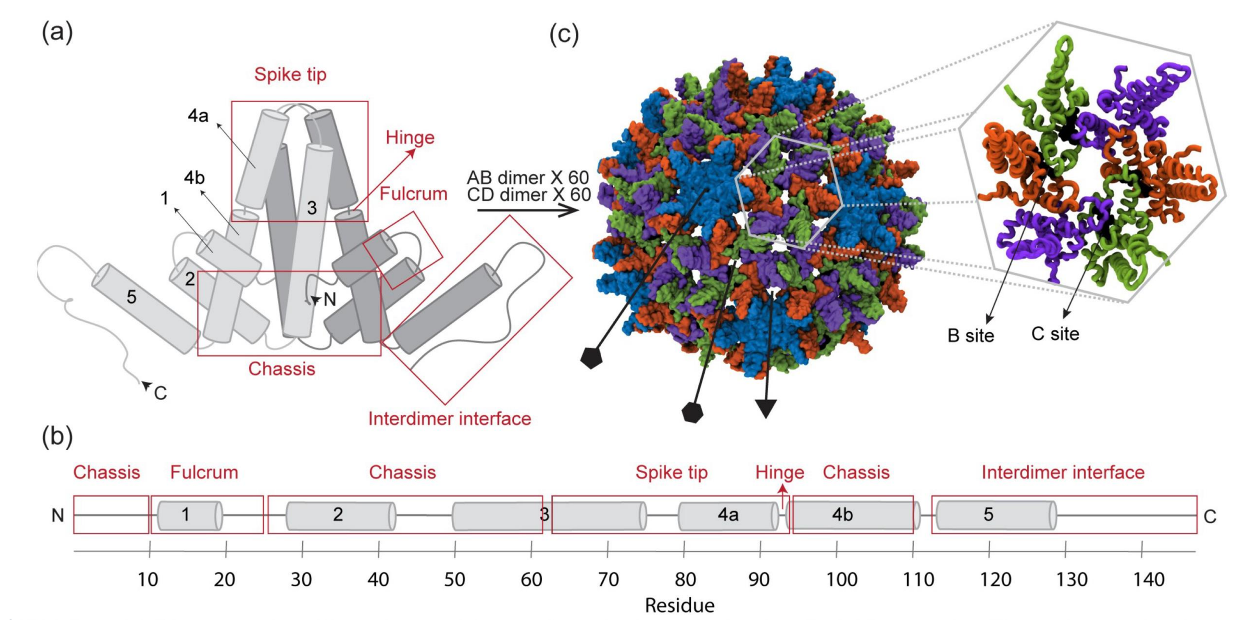
Introduction

Viruses use protein shells called capsids to protect their genomic data. These capsids assemble themselves spontaneously through hydrophobic and hydrophilic bonds, as well as electrostatic interactions.

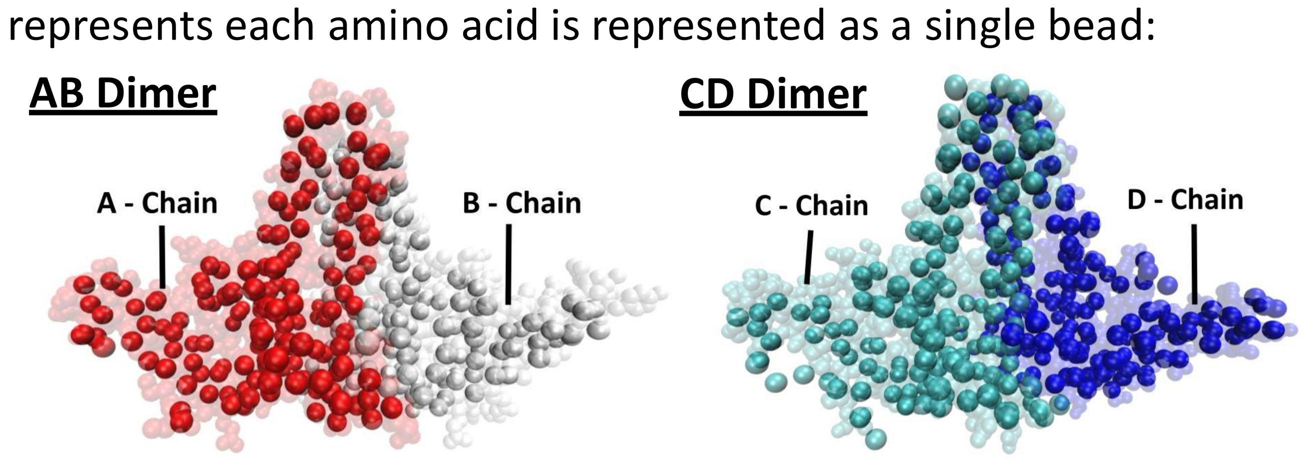


HBV Capsid Diagram detailing the residues of each domain in the Cp149 protein. The schematic also displays the assembled capsid and one of its vertices[1]

To understand detailed capsid assembly pathways, we aim to develop a computational model of the HBV capsid that is sufficiently tractable to simulate timescales relevant to capsid assembly while retaining atomic-level detail of the capsid structure

Residue-Based Model

To balance both computational costs and maintaining faithfulness to the structure of the protein at atomic resolution, our model represents each amino acid is represented as a single bead:



- Each chain is a different conformation of the same capsid protein
- The HBV capsid consists of 60 AB and 60 CD dimers

Comparing this to the data from atomistic simulation data we can see that the capsid stretching geometries by measuring dihedral angles in the capsid

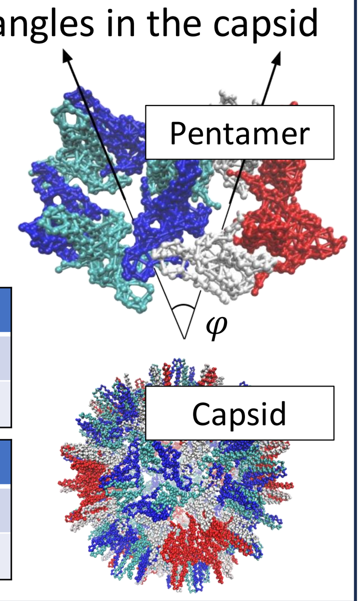
- The dihedral angle (φ) is an indicator of the flexibility of the intermediate
- Fully atomistic model and our residue-based model demonstrate a higher flexibility in a pentamer vs. a full capsid

Pentamer Dihedral

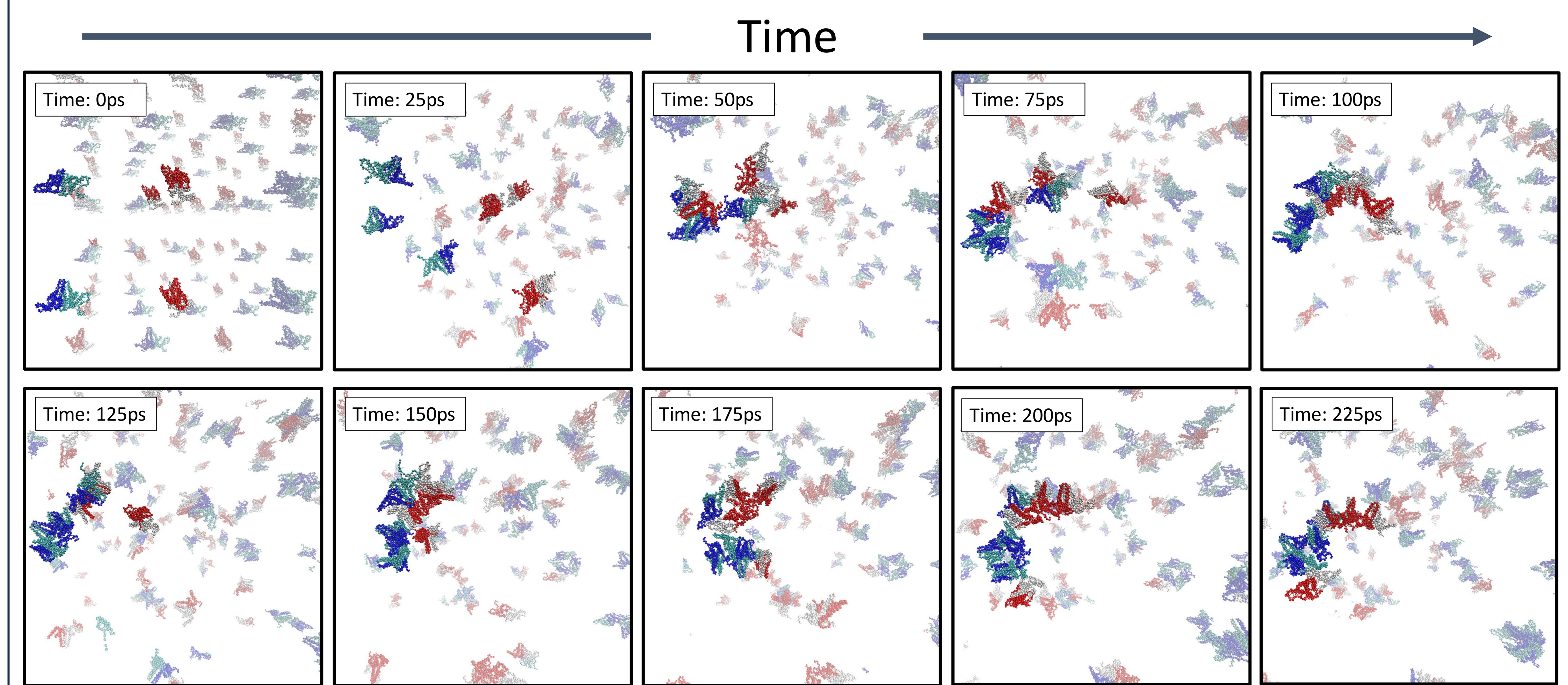
Dimer	Residue-Model	Atom-Model
AB	0.015	0.040
CD	0.014	0.040

Capsid Dihedral

Dimer	Residue-Model	Atom-Model
AB	0.208	0.094
CD	0.125	0.123



Analyzing Assembly



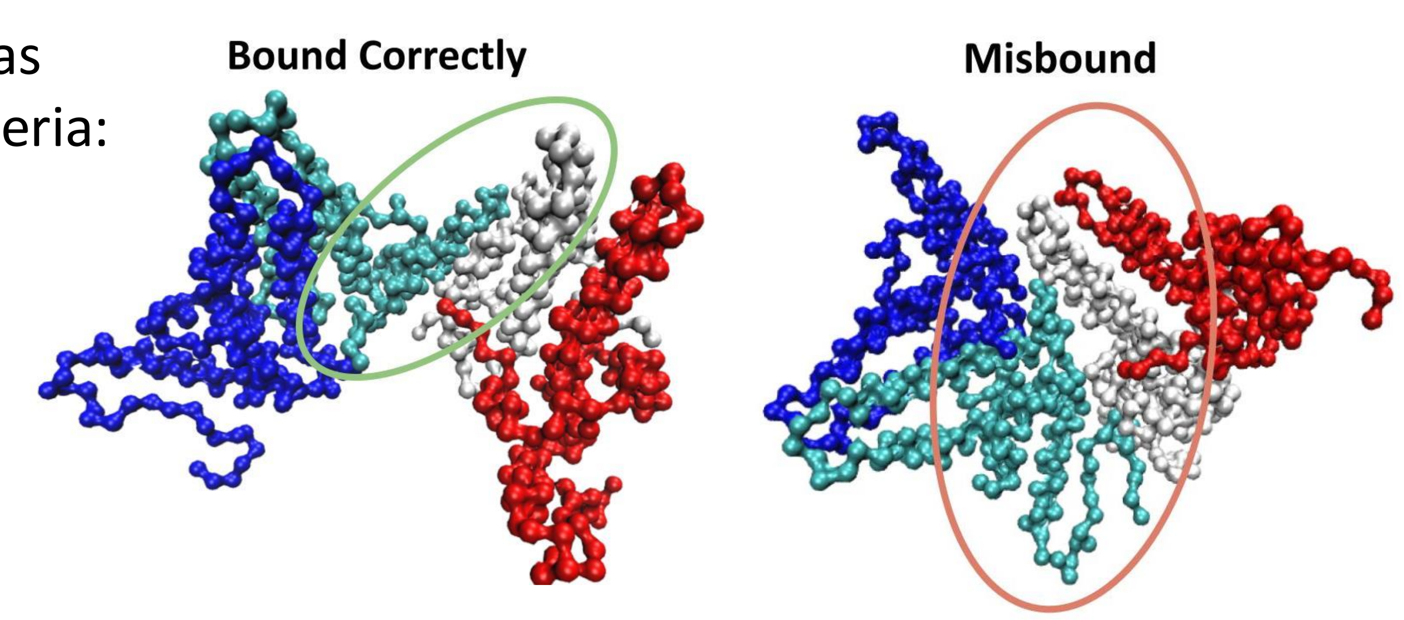
*1ps is 10^{-12}seconds

Malformed Assembly vs Correct Assembly

To measure whether a bond was made correctly we use two criteria:

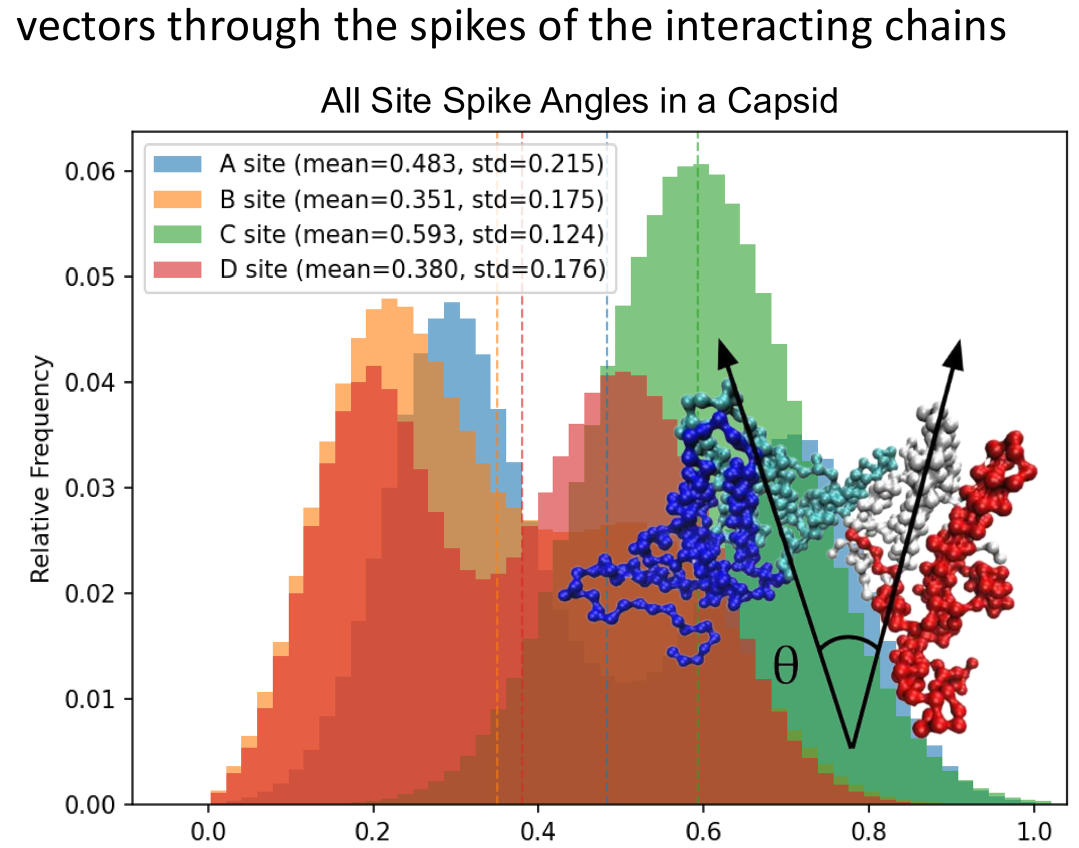
- Spike Angle
- Binding Angle

*Using data from the graphs below we set thresholds classification for well-bounded interfaces at **1.5 rad for the spike angle** and **1.5 rads for the binding angle**



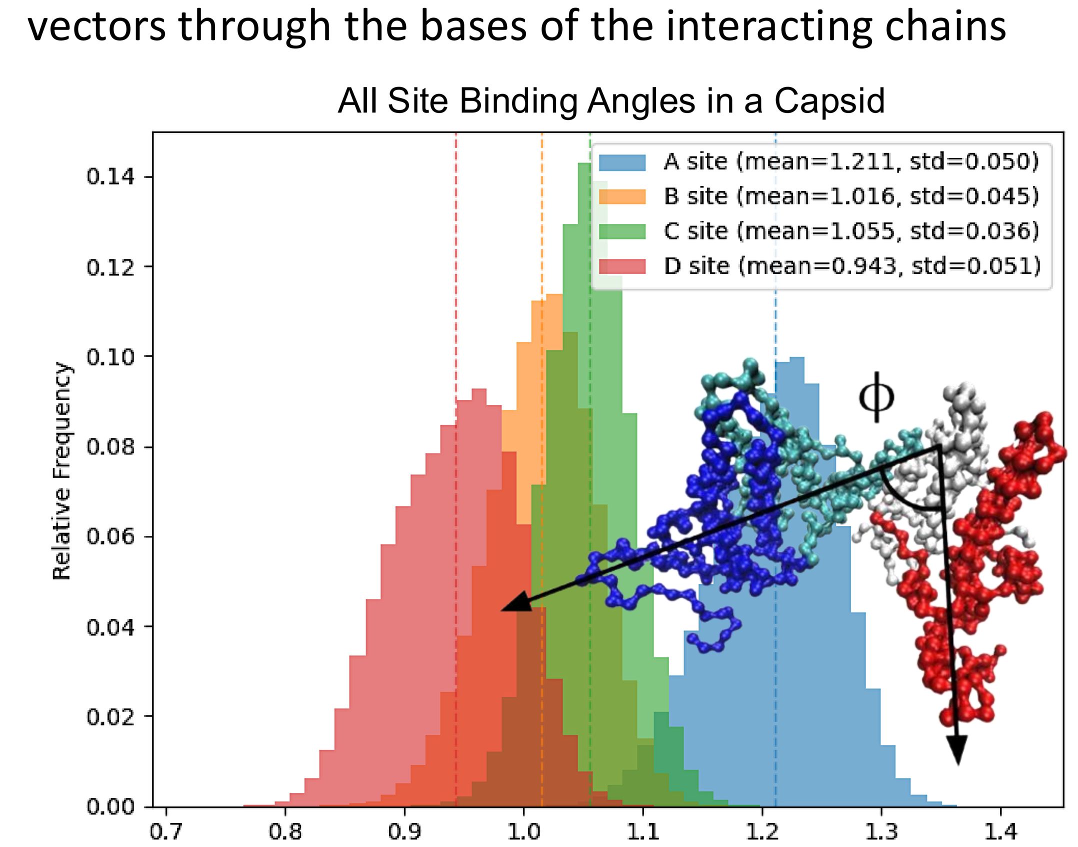
Spike Angle (θ)

The spike angle is the angle created by drawing two vectors through the spikes of the interacting chains



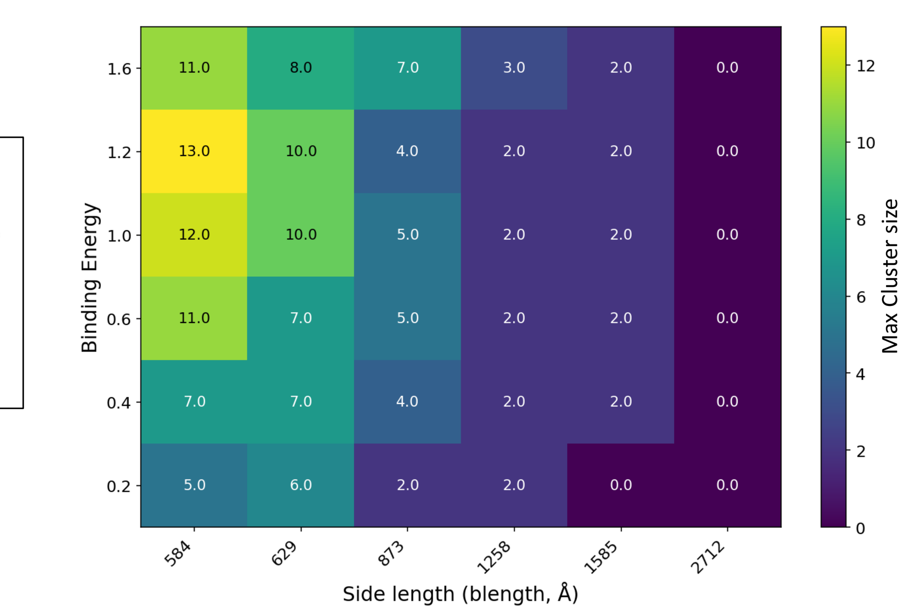
Binding Angle (ϕ)

The binding angle is the angle made by drawing two vectors through the bases of the interacting chains

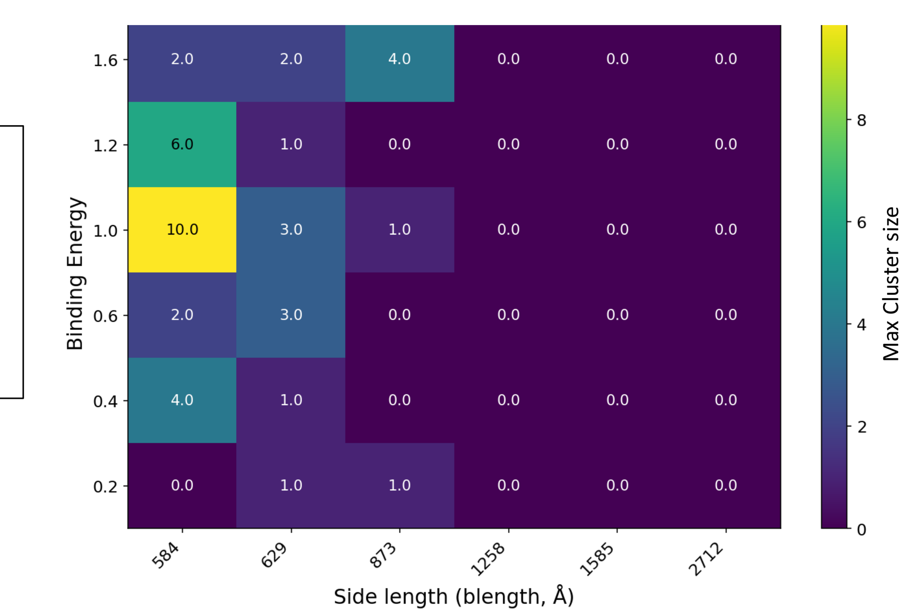


Conclusions

Total Cluster Size



Total Cluster Size - Well formed



- At high binding energies or high concentrations larger clusters start to form
- These clusters tend to have more malformed bonds that then struggle to disassociate
- By measuring the spike angle and binding angle of bonded interfaces we can detect malformed bonds

References

1. Pérez-Segura, C., Boon, C. G., & Hadden-Perilla, J. (2021). All-Atom MD Simulations of the HBV Capsid Complexed with AT130 Reveal Secondary and Tertiary Structural Changes and Mechanisms of Allostery. *Viruses*, *13*(4), 564. https://doi.org/10.3390/v13040564
2. Jason D Perlmutter, Cong Qiao, Michael F Hagan (2013) Viral genome structures are optimal for capsid assembly eLife 2:e00632. https://doi.org/10.7554/eLife.00632

Acknowledgments

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This research was completed at Brandeis University. The Brandeis campus sits on land that was sacred to the Massachusett nation, including four tribes existing today: the Mattakeeset, Natick, Ponkapoag, and Namasket. Both Native Americans and Africans were enslaved in the colony of Massachusetts.